

# José Manuel Requena Plens

## Firmware & Software Engineer

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On-site, remote or hybrid

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## Summary

**Embedded firmware and software engineer** with experience building industrial firmware for solar inverters (C, STM32, FreeRTOS) and a solid **R&D** background. Focused on low-level optimization, industrial communications (Modbus) and hardware test automation. I maintain *open source* projects used by developers worldwide. Targeting **firmware, software and QA** roles, bringing engineering rigor and a focus on quality.

## Experience

### R&D Software Developer

Apr 2023 - May 2026

Power Electronics — Solar + Storage

Lliria, Spain

- Reduced RTOS task stack usage by 30%–50%, optimizing memory on STM32 microcontrollers.
- Optimized Modbus TCP request handling, cutting CPU usage under load (40,000 req/s) from 80% to 17% with no dropped responses.
- Redesigned embedded file-system access, cutting access times by over 50%.
- Implemented a scheduled purge (daily/monthly) that safely clears thousands of files, preventing storage exhaustion under continuous operation.
- Built and maintained industrial communications in C with hardware abstraction layers (HAL): Modbus RTU/TCP, UART, I2C and SPI.
- Automated the test suite: ~50–80 unit tests per module (Ceedling, thousands in total) reaching 100% line and branch coverage via standard-library mocks/fakes, plus hundreds of on-hardware integration tests (pytest) — TCP/FTP performance and FTP security —; all in CI.
- Reviewed merge requests across a 12-person team, catching bugs and proposing improvements before integration.

### Predoctoral researcher

Jul 2021 - Jul 2022

UPV · CSIC · i3M

Valencia, Spain

- Designed acoustic metamaterials (metadiffusers and lenses) that reduce the footprint of conventional diffusers, validated with MATLAB and COMSOL simulations.
- Manufactured and characterized prototypes; published results at international conferences (Euronoise, Tecniacústica).
- Supervised a Master's thesis in Acoustic Engineering (UPV), from research design to defense.

### Researcher in European project

Feb 2020 - Jul 2021

UPV · European Space Agency (ESA) — IGIC

Gandia, Spain

- Designed and experimentally validated a noise-mitigation metamaterial for the VEGA rocket launch (ESA project AO/1-9479/18/NL/LvH), meeting 100% of project objectives with results accepted by ESA.
- Developed acoustic measurement software per ISO 10534-2 and ASTM 2611 ([A|Lab](#)) and a 3-axis robotic system for experimental measurements.
- Co-author of international publications/talks (Euronoise, ECSSMET).

## Technical Skills

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**Embedded Firmware & Electronics:** C, STM32 & ARM, ESP32 / ESP-IDF, PlatformIO, Texas Instruments, FreeRTOS / RTOS, Comm. Protocols, HW Debugging, Keil uVision, Instrumentation, Secure Boot & Embedded Security, Doxygen

**Software Engineering & Tools:** Python, C++, Go, TypeScript, Bash, Clang Tooling, Git, GitHub, GitLab, Docker, MATLAB, CMake, JetBrains Toolbox

**Quality, CI/CD & DevSecOps:** GitHub Actions, GitLab CI, Jenkins, SonarQube / SonarCloud, GoReleaser, Static Analysis & SAST, Testing, pre-commit hooks

**Networking:** TCP/IP & Networking, MikroTik RouterOS, WireGuard / VPN, IPv6 & Dual-stack, QUIC / HTTP3, DNS / Dynamic DNS

**Security & Cryptography:** Applied Cryptography, TLS / mTLS / PKI, Nginx Hardening, CrowdSec, Network Hardening

**AI & Agentic Engineering:** Model Context Protocol (MCP), AI Coding Agents, LLM Integration

**Simulation & Acoustics:** COMSOL Multiphysics, LabView, Acoustic Instrumentation

**General & Platforms:** Linux / Unix, MacOS, Windows, LaTeX, HTML / CSS

## Open Source Projects

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**gitlab-mcp-server** — Go · Model Context Protocol server

27 ★ · 35 releases · ~30k downloads · [Listed on Glama](#), [mcp.so](#), [Cursor](#) & [mcpservers.org](#)

Open-source GitLab MCP server that exposes 1,000+ GitLab operations to AI assistants through a dynamic 2-tool find/execute design, with stdio/HTTP/OAuth transports and safe & read-only modes.

**PyOctaveBand** — Python · Signal processing

107 ★ · 22 releases · [PyPI](#)

Octave-band and fractional octave-band filtering library for time-domain signals, published on PyPI.

**cs-routeros-bouncer** — Go · Firewall · MikroTik · WAF

10 ★ · 14 releases · 316 downloads · [On the CrowdSec Hub](#)

CrowdSec bouncer for MikroTik RouterOS: feeds the collaborative IPS/WAF CrowdSec decisions into the RouterOS firewall (address lists) through its API, blocking malicious IPs in real time.

**Cloudflare-DNS-Updater** — Bash · DevOps

26 ★ · 5 releases · 45 downloads

Dynamic DNS client for Cloudflare (IPv4/IPv6) written in Bash, with a bats test suite and coverage reporting.

## Education

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**PhD in Health and Wellness Technologies**

2023

*Polytechnic University of Valencia (i3M / UMIL)*

*Valencia, Spain*

Research on acoustic metamaterials for medical imaging and ultrasound beam control (programme not completed; redirected to industrial software development).

**Master's Degree in Acoustical Engineering**

2019

*Polytechnic University of Valencia (EPSG)*

*Gandia, Spain*

Graduated with honors; Master's thesis awarded a special distinction.

**Degree in Telecommunication Engineering in Sound and Image**

2018

*University of Alicante (EPS)*

*Alicante, Spain*

Two specializations: Acoustic Engineering and Audiovisual Technology.

**Higher Vocational Degree — Superior Sound Technician**

2011

*IES Luis García Berlanga*

*Alicante, Spain*

**Intermediate Vocational Degree — Electronics Technician**

2009

*IES Salesianos (San José Artesano)*

*Elche, Spain*